Jonathan Liu

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EDUCATION

University of Washington — **Seattle, WA**
Bachelor of Science in Electrical and Computer Engineering | GPA: 3.93 — *Sep 2024 – June 2027*

- **Honors:** IEEE HKN Honor Society (Top 8% ECE), Tau Beta Pi Engineering Honor Society (Top 12.5%).
- **Relevant Coursework:** Analog IC Design, Microelectronic Circuits, Digital Circuit Design, Signals & Systems, Power Electronics, Control Systems, Computer Architecture, Data Structures, Hardware/Software Interface.

Tohoku University STEM Summer Program — **Sendai, Japan**
Nanoelectronics Fabrication & Robotics Immersion | GPA: 4.0 — *June – July 2025*

SKILLS

VLSI: Cadence Virtuoso (Schematic & Layout), **ADE Explorer**, **ADE Assembler**, **GPDK045** (45nm CMOS), **LTspice**, SPICE, Op-Amps, OTAs, Current Mirrors, Differential Pairs, Frequency Compensation.
Hardware & Embedded: Altium Designer, KiCad, STM32 (CubeIDE/CubeMX/HAL), Arduino, FPGA (SystemVerilog), Intel Quartus, ModelSim, BLDC Motor Control, LDOs, Soldering, Oscilloscopes, Logic Analyzers.
Software & Simulation: C/C++, Python (NumPy/SciPy/PyTorch), Java, MATLAB, Fusion360, OpenSCAD, Linux/Bash, Git.
Languages: English, Chinese (Mandarin & Cantonese), Japanese.

RELEVANT EXPERIENCE

Undergraduate Research Assistant (Embedded Systems) — **Jan 2026 – Present**
Sensors, Energy, and Automation Laboratory (SEAL) — *Seattle, WA*

- Developed **random forest** classification models to analyze **PPG biosignals** and detect fatigue levels from heart biometrics in safety-critical environments.
- Designed custom **3D-printed PCB mounts** using **OpenSCAD** for wearable body sensor placement and integration.

Lead (Nov 2025 – Present) | Drone Hardware Engineer — **Oct 2024 – Present**
Husky Robotics Drone Subteam — *Seattle, WA*

- **Technical Lead:** Spearhead recruitment and technical onboarding for 20+ members; manage the full development lifecycle of three custom **PCBs** (Flight Controller, ESC, PDB), BOM selection, and Professional Design Reviews.
- **Custom Flight Controller V0.1:** Architecting and laying out a Pixhawk-class flight controller from scratch in **Altium Designer** around an **STM32H743VGT6**; integrated **INA219** current/power telemetry, CP2102N USB-UART debug bridge, ESD-protected USB-C, and full **UART/CAN/SPI/I^{2}C/Ethernet** routing.
- **Embedded Firmware:** Writing bare-metal peripheral drivers in **STM32CubeIDE/CubeMX** using the **STM32 HAL** for SPI/UART/I^{2}C/CAN-FD/USB stacks, enabling **DSHOT600** motor control, **MAVLink** telemetry, and **PX4**-compatible sensor fusion on a **Jetson Orin Nano** + Pixhawk 6C companion-computer bridge.
- **Testing:** Developed an automated thrust stand with hardware-in-the-loop (**HIL**) fixtures to characterize **BLDC** motor performance via oscilloscopes, optimizing propulsion efficiency by 25%.

Autonomous Systems Engineering Intern — **July – Sep 2025**
ARIDGE (XPeng AEROHT) — *Guangzhou, China*

- Wrote **Python** scripts to train **computer vision** models for autonomous landing zone classification and document categorization.
- Validated **LiDAR** point cloud and imagery data to verify **sensor fusion** integrity for autonomous navigation pipelines; managed logging workflows and cross-functional technical communication across engineering teams.

PROJECTS

5T-OTA-Based CMOS Operational Amplifier (GPDK045) — *May 2026*

- Designed and optimized a **CMOS op-amp** built around a **5T OTA** input core in Cadence **GPDK045** (45nm) using **Cadence Virtuoso** (**Schematic & Layout**); hand-sized the differential pair, current-mirror load, tail source, and output stage from first-principles small-signal analysis, then ran **AC/DC/transient** sweeps in **ADE Explorer** and applied **ADE Assembler** Brent-Powell optimization (all-saturation constraints) to achieve **13.7 dB** gain, **629 MHz** bandwidth, **3.05 GHz** GBW, and 0.89 V swing on a 1 V supply.

Adjustable AC-DC Boost Converter — *May 2026*

- Designed, simulated in **LTspice**, and fully **breadboarded** a two-stage **AC-DC** converter stepping $\pm$7.5 VAC to adjustable **+10–20 VDC** ($<$100 mV ripple) via full-wave bridge rectification, NE555-clocked (100 kHz) 2N7000 **boost** stage, and op-amp/Zener feedback regulation.

Three-Band Active Audio Equalizer — *March 2026*

- Designed and constructed a three-band active audio equalizer using the **LM348** op-amp; derived theoretical transfer functions and verified frequency response via full-system **LTspice** simulation.

FPGARhythm — *March 2026*

- Developed a hardware-based rhythm game in **SystemVerilog** on an Intel FPGA, processing user inputs and rendering cascading notes via **VGA** output.

ORGANIZATIONS & AWARDS

- **Dean's List** – UW (2024 – Present) | **UW Line-Following Car** – All-Time Fastest Record (Feb 2026)
- **Events Officer** – IEEE UW (Jan 2026 – Present) | Hong Kong Student Association at UW (Oct 2025 – Present)